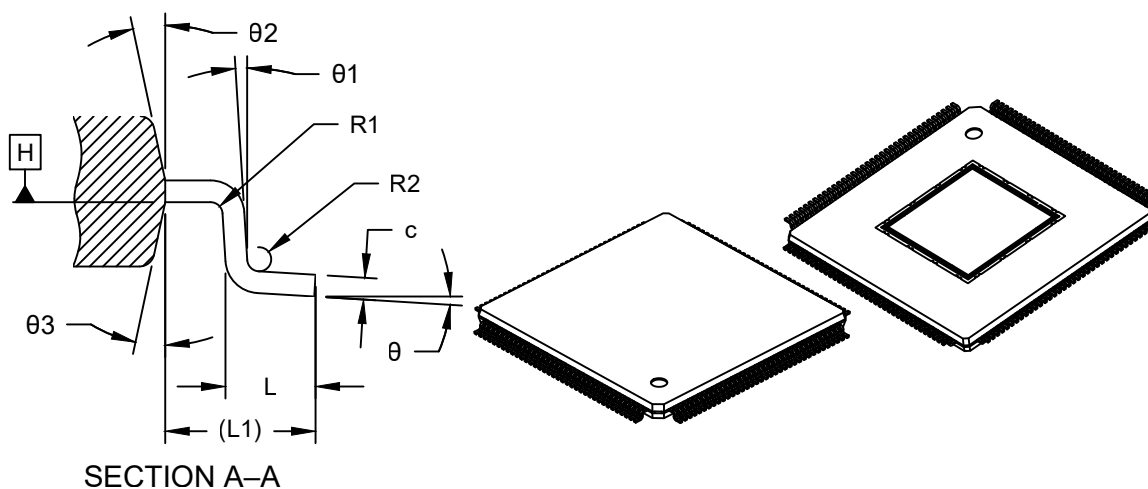




## Packaging Diagrams and Parameters

### 144-Lead Plastic Thin Quad Flat Pack (8XW) - 16x16x1.0mm Body [TQFP] With 8x8mm Grooved Exposed Pad

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



| Units                    |    | MILLIMETERS |      |      |
|--------------------------|----|-------------|------|------|
| Dimension Limits         |    | MIN         | NOM  | MAX  |
| Number of Terminals      | N  | 144         |      |      |
| Pitch                    | e  | 0.40 BSC    |      |      |
| Overall Height           | A  | -           | -    | 1.20 |
| Standoff                 | A1 | 0.05        | -    | 0.15 |
| Molded Package Thickness | A2 | 0.95        | 1.00 | 1.05 |
| Overall Length           | D  | 18.00 BSC   |      |      |
| Molded Package Length    | D1 | 16.00 BSC   |      |      |
| Exposed Pad Length       | D2 | 8.00 BSC    |      |      |
| Overall Width            | E  | 18.00 BSC   |      |      |
| Molded Package Width     | E1 | 16.00 BSC   |      |      |
| Exposed Pad Width        | E2 | 8.00 BSC    |      |      |
| Terminal Width           | b  | 0.13        | 0.16 | 0.23 |
| Terminal Thickness       | c  | 0.09        | -    | 0.20 |
| Terminal Length          | L  | 0.45        | 0.60 | 0.75 |
| Footprint                | L1 | 1.00 REF    |      |      |
| Lead Bend Radius         | R1 | 0.08        | -    | -    |
| Lead Bend Radius         | R2 | 0.08        | -    | 0.20 |
| Foot Angle               | θ  | 0°          | 3.5° | 7°   |
| Lead Angle               | θ1 | 0°          | -    | -    |
| Mold Draft Angle         | θ2 | 11°         | 12°  | 13°  |
| Mold Draft Angle         | θ3 | 11°         | 12°  | 13°  |

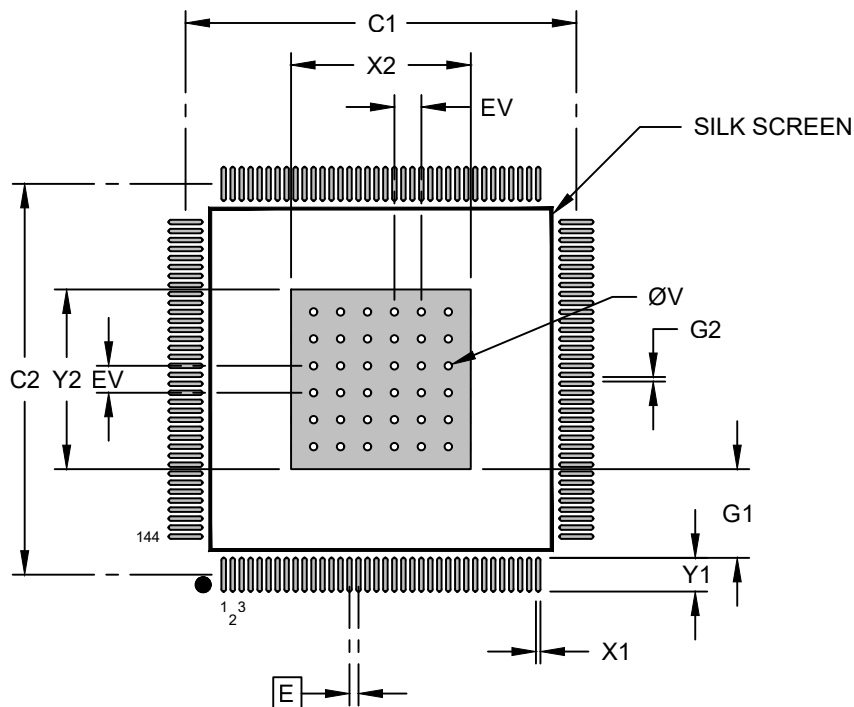
**Notes:**

- The Pin 1 visual index feature may vary, but it must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.  
REF: Reference Dimension, usually without tolerance, for information purposes only.

## Land Pattern

### 144-Lead Plastic Thin Quad Flat Pack (8XW) - 16x16x1.0mm Body [TQFP] With 8x8mm Grooved Exposed Pad

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



### RECOMMENDED LAND PATTERN

| Units                            |    | MILLIMETERS |       |      |
|----------------------------------|----|-------------|-------|------|
| Dimension Limits                 |    | MIN         | NOM   | MAX  |
| Contact Pitch                    | E  | 0.40 BSC    |       |      |
| Optional Center Pad Width        | X2 |             |       | 8.00 |
| Optional Center Pad Length       | Y2 |             |       | 8.00 |
| Contact Pad Spacing              | C1 |             | 17.40 |      |
| Contact Pad Spacing              | C2 |             | 17.40 |      |
| Contact Pad Width (Xnn)          | X1 |             |       | 0.20 |
| Contact Pad Length (Xnn)         | Y1 |             |       | 1.50 |
| Contact Pad to Center Pad (Xnn)  | G1 | 3.95        |       |      |
| Contact Pad to Contact Pad (Xnn) | G2 | 0.20        |       |      |
| Thermal Via Diameter             | V  |             | 0.33  |      |
| Thermal Via Pitch                | EV |             | 1.20  |      |

**Notes:**

- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-02676 Rev A